

TTK4240 Industriell elektroteknikk

January 07, 2025

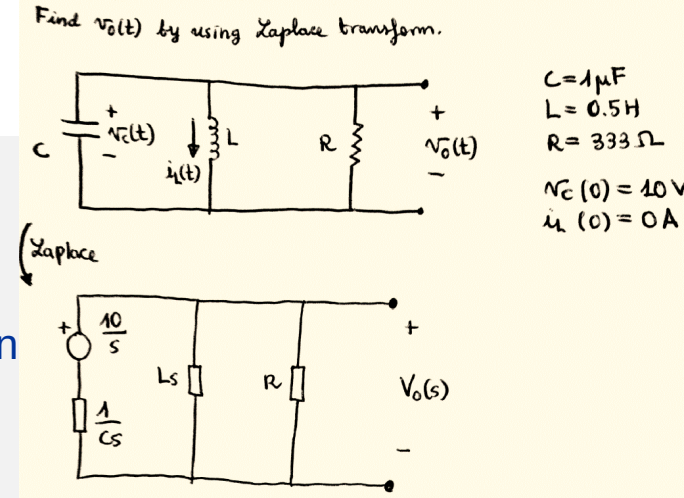
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Lecture - Week 02 Part II

Laplace Transform in Circuit Analysis

- Circuit elements in «s» domain
 - Resistor, Inductor and Capacitor in «s» domain
- Partial fraction expansion
- Circuit Analysis in «s» domain
 - Natural response of an RC circuit
 - Step response of a parallel circuit: constant-value input
 - Step response of a parallel circuit: sinusoidal input
- The Transfer Function
- Use of the Thevenin's Equivalent



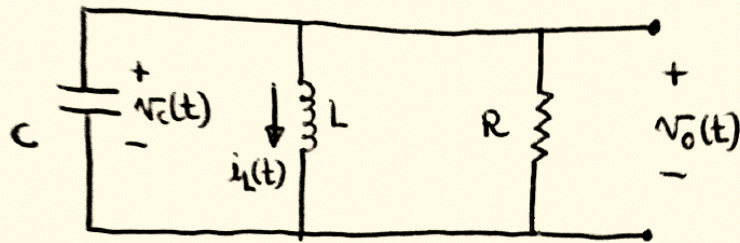
Chapter 13 Nilsson (all except 13.6 and 13.8)

The LAPLACE TRANSFORM

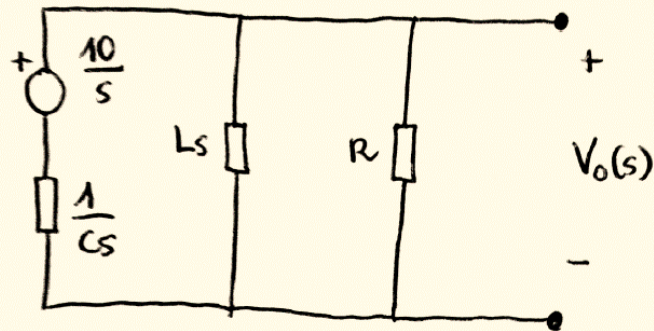
$$F(s) = \mathcal{L}[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$$

keypoint

Find $v_o(t)$ by using Laplace transform.



(Laplace



$$\begin{aligned}C &= 1\mu\text{F} \\L &= 0.5\text{H} \\R &= 333\Omega\end{aligned}$$

$$\begin{aligned}v_c(0) &= 10\text{V} \\i_L(0) &= 0\text{A}\end{aligned}$$

Transforms ordinary differential Equations (ODE) into algebraic Equations that are more easy to manipulate to find a solution

Why Laplace Transform?

Time domain

$h(t)$

Solution is difficult

$u(t)$



System described by ODE



$y(t)$

$$= \int_0^t h(t - \tau) u(\tau) d\tau$$

The complete response of a circuit

KEYPOINT

- **Important:** the complete response of a circuit to a sinusoidal source has two parts:

Complete response = steady-state response (as the same form as the input) + transient response (decaying to zero)

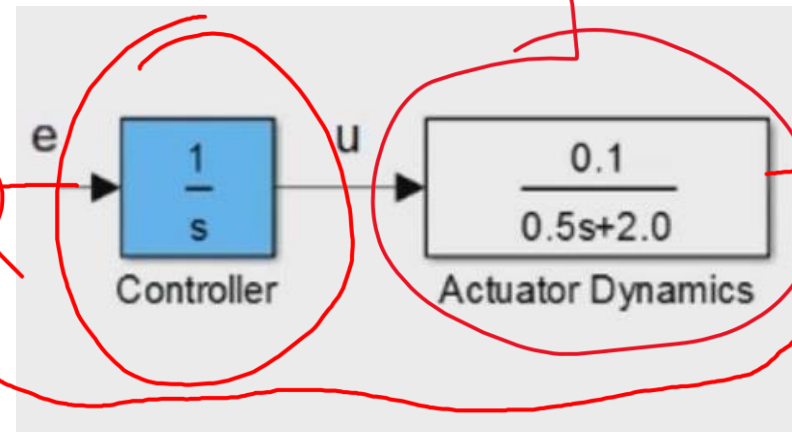
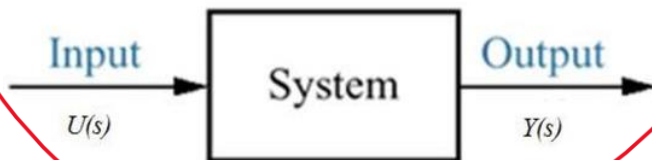
- **When designing a circuit**, it is important to be able to *determine the complete response of a circuit*. The Laplace transform techniques can be used to find the complete response of a circuit to a sinusoidal source.

Why?

Example of use

- Used to introduce the concept of the *transfer function* as a tool for analyzing the *steady-state sinusoidal response* of a circuit when the frequency of the sinusoidal source is varied

$$G(s) = \frac{Y(s)}{U(s)}$$

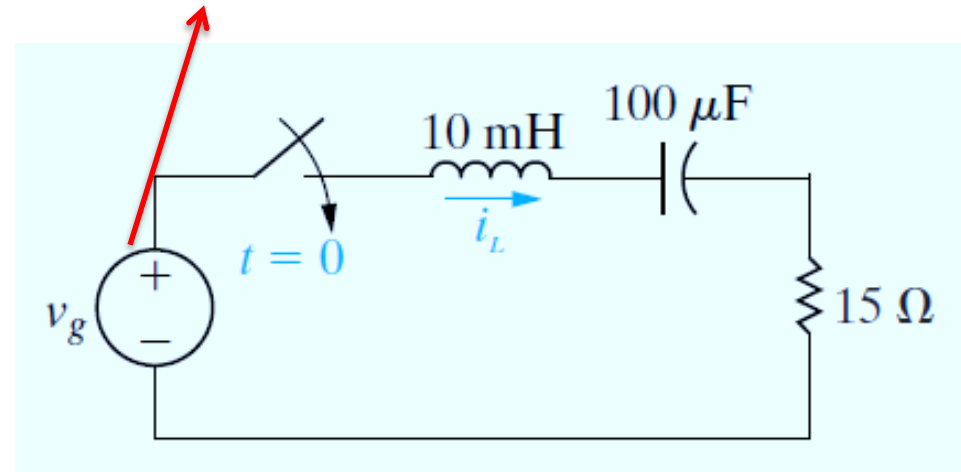


Use in Circuit Analysis

Excitation sources with Laplace transform

Laplace transform transforms a set of *integrodifferential equations* from the *time domain* to a set of *algebraic equations* in the *frequency domain*.

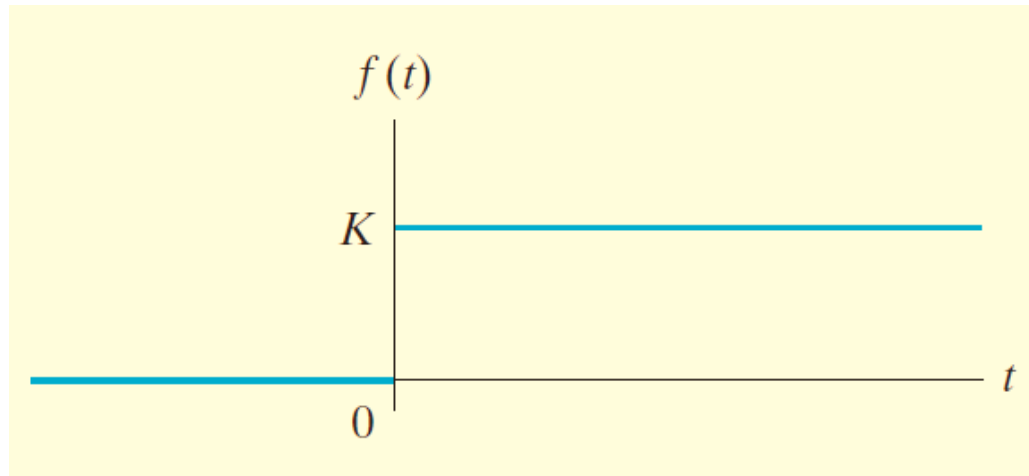
$$v_g = L \frac{di}{dt} + \frac{1}{c} \int i dt + iR$$



In *linear circuit analysis*, we excite circuits with *sources that have Laplace transforms*. Excitation functions which do not have Laplace transforms, are of no interest here.

The Step Function

Review



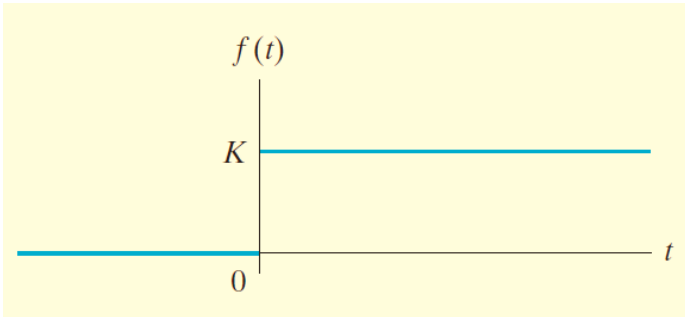
Function with discontinuity
or jump at the origin:
Zero for $t < 0$ and K for $t > 0$

$$\begin{aligned} Ku(t) &= 0, & t < 0, \\ Ku(t) &= K, & t > 0. \end{aligned}$$

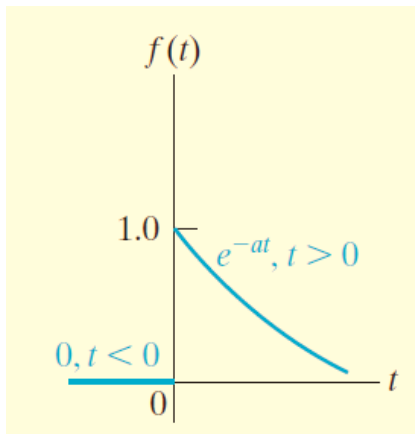
Functional Transforms

Review

Step Function



$$\begin{aligned}\mathcal{L}\{u(t)\} &= \int_{0^-}^{\infty} f(t)e^{-st} dt = \int_{0^+}^{\infty} 1e^{-st} dt \\ &= \left. \frac{e^{-st}}{-s} \right|_{0^+}^{\infty} = \frac{1}{s}.\end{aligned}$$



Exponential Function

$$\mathcal{L}\{e^{-at}\} = \int_{0^+}^{\infty} e^{-at} e^{-st} dt = \int_{0^+}^{\infty} e^{-(a+s)t} dt = \frac{1}{s+a}.$$

Laplace Transform of Sinusoid

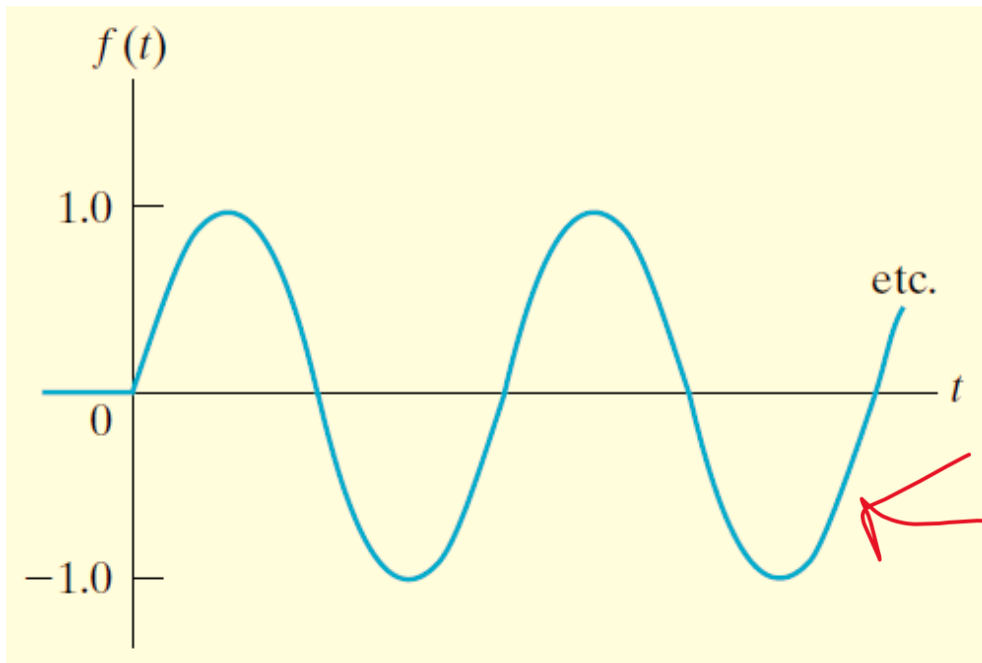
Review

$$\mathcal{L}\{\sin \omega t\} = \int_{0^-}^{\infty} (\sin \omega t) e^{-st} dt = \int_{0^-}^{\infty} \left(\frac{e^{j\omega t} - e^{-j\omega t}}{2j} \right) e^{-st} dt$$

$$= \int_{0^-}^{\infty} \frac{e^{-(s-j\omega)t} - e^{-(s+j\omega)t}}{2j} dt$$

$$= \frac{1}{2j} \left(\frac{1}{s - j\omega} - \frac{1}{s + j\omega} \right)$$

$$= \frac{\omega}{s^2 + \omega^2}$$



Type	$f(t)$ ($t > 0-$)	$F(s)$
(impulse)	$\delta(t)$	1
(step)	$u(t)$	$\frac{1}{s}$
(ramp)	t	$\frac{1}{s^2}$
(exponential)	e^{-at}	$\frac{1}{s + a}$
(sine)	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
(cosine)	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
(damped ramp)	te^{-at}	$\frac{1}{(s + a)^2}$
(damped sine)	$e^{-at} \sin \omega t$	$\frac{\omega}{(s + a)^2 + \omega^2}$
(damped cosine)	$e^{-at} \cos \omega t$	$\frac{s + a}{(s + a)^2 + \omega^2}$

Operational Transforms

Review

$$\mathcal{L}\{Kf(t)\} = KF(s).$$

Multiplication by a constant

$$\mathcal{L}\{f_1(t) + f_2(t) - f_3(t)\} = F_1(s) + F_2(s) - F_3(s),$$

Addition, Substraction

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = sF(s) - f(0^-),$$

Differentiation

$$\begin{aligned}\mathcal{L}\left\{\frac{d^n f(t)}{dt^n}\right\} &= s^n F(s) - s^{n-1}f(0^-) - s^{n-2}\frac{df(0^-)}{dt} \\ &\quad - s^{n-3}\frac{d^2 f(0^-)}{dt^2} - \dots - \frac{d^{n-1} f(0^-)}{dt^{n-1}}.\end{aligned}$$

very good!

Operational Transforms

Review

$$\mathcal{L} \left\{ \int_{0^-}^t f(x) dx \right\} = \frac{F(s)}{s},$$

very
used!

Integration

$$\mathcal{L}\{f(t - a)u(t - a)\} = e^{-as}F(s), \quad a > 0.$$

Translation in time domain

$$\mathcal{L}\{e^{-at}f(t)\} = F(s + a),$$

Translation in frequency domain

$$\mathcal{L}\{f(at)\} = \frac{1}{a}F\left(\frac{s}{a}\right), \quad a > 0,$$

Scale changing

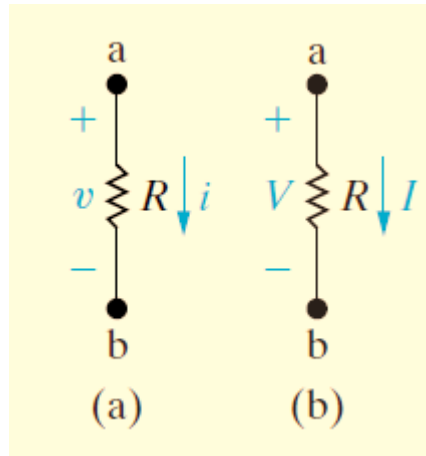
Most used Laplace Transforms

Review

Operation	$f(t)$	$F(s)$
Multiplication by a constant	$Kf(t)$	$KF(s)$
Addition/subtraction	$f_1(t) + f_2(t) - f_3(t) + \dots$	$F_1(s) + F_2(s) - F_3(s) + \dots$
First derivative (time)	$\frac{df(t)}{dt}$	$sF(s) - f(0^-)$
Second derivative (time)	$\frac{d^2f(t)}{dt^2}$	$s^2F(s) - sf(0^-) - \frac{df(0^-)}{dt}$
n th derivative (time)	$\frac{d^n f(t)}{dt^n}$	$s^n F(s) - s^{n-1}f(0^-) - s^{n-2}\frac{df(0^-)}{dt}$ $- s^{n-3}\frac{df^2(0^-)}{dt^2} - \dots - \frac{d^{n-1}f(0^-)}{dt^{n-1}}$
Time integral	$\int_0^t f(x) dx$	$\frac{F(s)}{s}$
Translation in time	$f(t - a)u(t - a), a > 0$	$e^{-as}F(s)$
Translation in frequency	$e^{-at}f(t)$	$F(s + a)$
Scale changing	$f(at), a > 0$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
First derivative (s)	$tf(t)$	$-\frac{dF(s)}{ds}$
n th derivative (s)	$t^n f(t)$	$(-1)^n \frac{d^n F(s)}{ds^n}$
s integral	$\frac{f(t)}{t}$	$\int_s^\infty F(u) du$

Circuit Elements in s -domain

Circuit elements in the «s» domain



Hand notes

A resistor in the «s» domain

$$v = Ri.$$

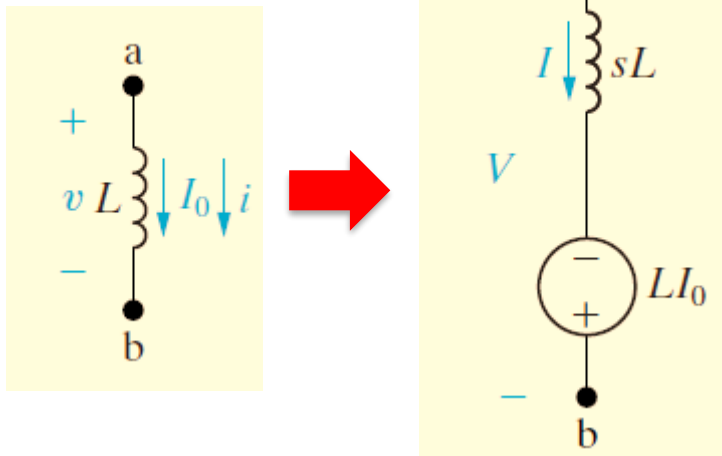
$$V = RI,$$

$$V = \mathcal{L}\{v\} \quad \text{and} \quad I = \mathcal{L}\{i\}.$$

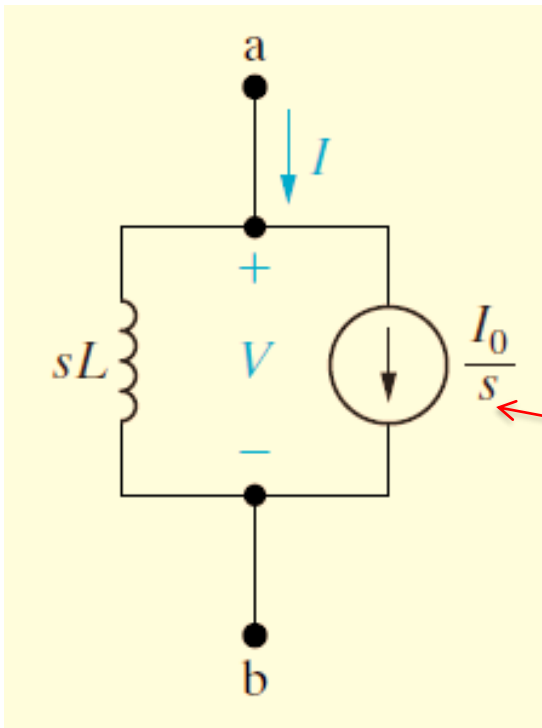
An inductor in the «s» domain

$$v = L \frac{di}{dt}.$$

$$V = L[sI - i(0^-)] = sLI - LI_0.$$



Inductor: solving for the current

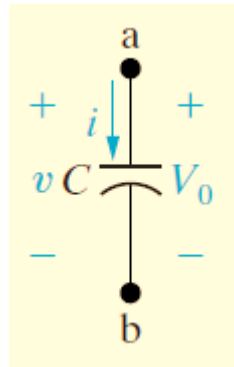


Applying Laplace and solving for the current the initial conditions are appearing

$$V = L[sI - i(0^-)] = sLI - LI_0.$$

$$I = \frac{V + LI_0}{sL} = \frac{V}{sL} + \frac{I_0}{s}.$$

Capacitor in the «s» domain



$$i = C \frac{dv}{dt}.$$

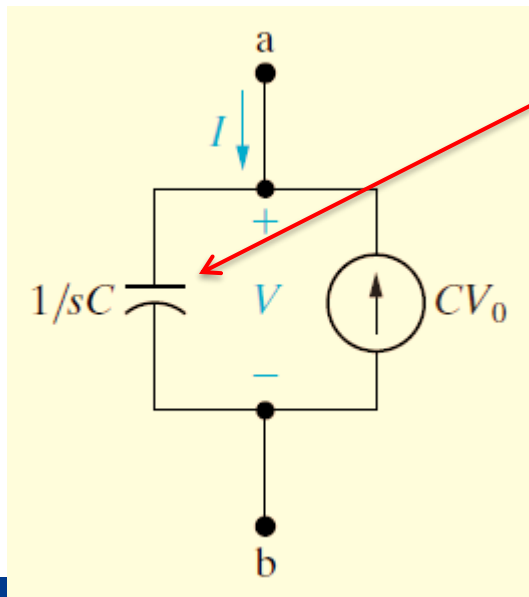
Solving for the current /

$$I = C[sV - v(0^-)]$$

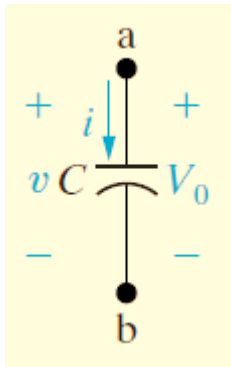
Voltage divided
by the impedance

$$I = sCV - CV_0,$$

$$V = \left(\frac{1}{sC} \right) I + \frac{V_0}{s}.$$



Capacitor in the «s» domain



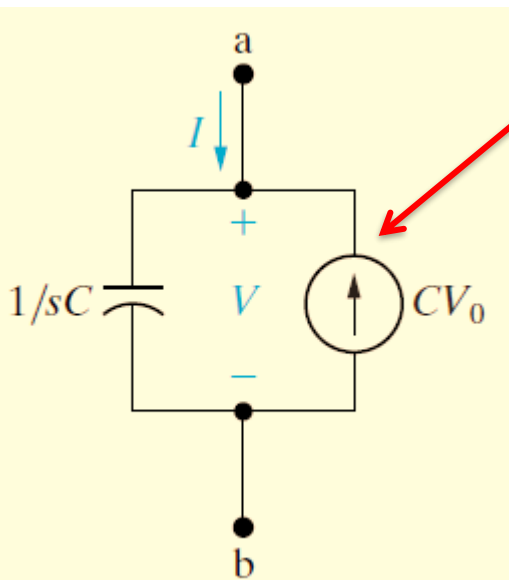
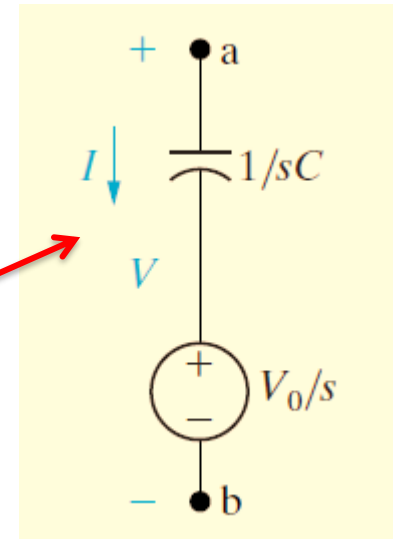
$$i = C \frac{dv}{dt}$$

Solving for the voltage V

$$I = C[sV - v(0^-)]$$

$$I = sCV - CV_0,$$

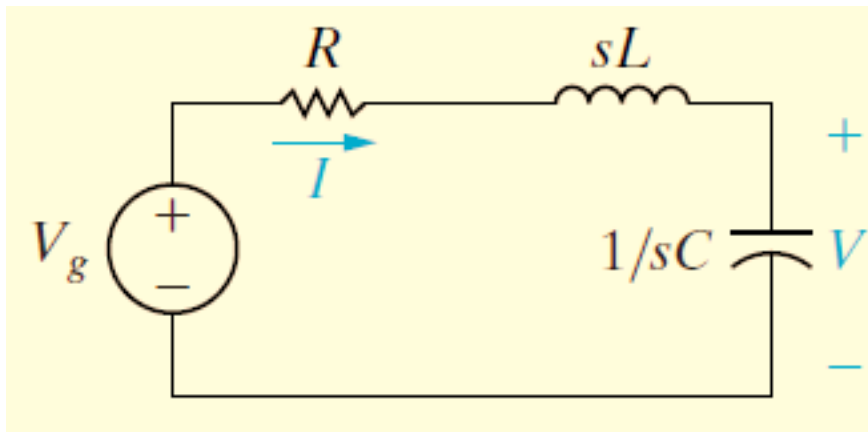
$$V = \left(\frac{1}{sC} \right) I + \frac{V_0}{s}$$



Ohm's Law in the «s» domain

$$V = ZI,$$

Z refers to the s-domain impedance



$$\text{alg } \sum I = 0,$$

$$\text{alg } \sum V = 0.$$

Kirchhoff's Laws
equivalent in s-domain

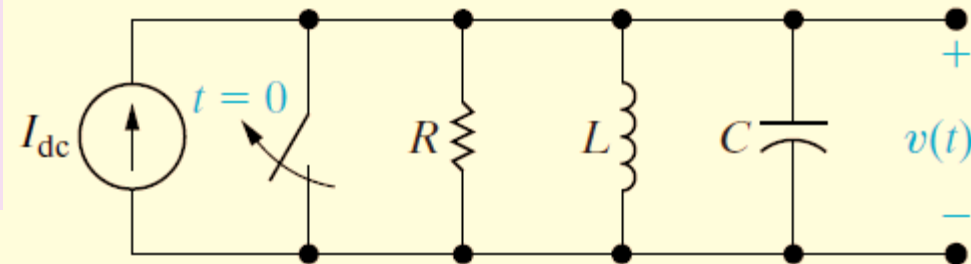
Applying the Laplace Transform

Step
f(t)

- Linear circuit with lumped constant parameters
- **No initial energy is stored** in the circuit at the instant when the switch is opened.
- The problem is to find the time-domain expression for $v(t)$ when $t \geq 0$

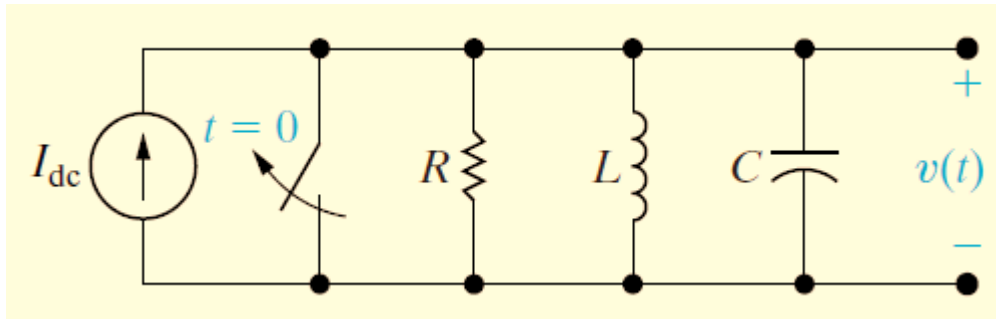
$$\frac{v(t)}{R} + \frac{1}{L} \int_0^t v(x) dx + C \frac{dv(t)}{dt} = I_{dc} u(t).$$

$$\frac{V(s)}{R} + \frac{1}{L} \frac{V(s)}{s} + C[sV(s) - v(0^-)] = I_{dc} \left(\frac{1}{s} \right),$$



Applying the Laplace Transform

Review



Problem:
solve for $v(t)$?

$$\frac{v(t)}{R} + \frac{1}{L} \int_0^t v(x) dx + C \frac{dv(t)}{dt} = I_{dc} u(t).$$

time-domain equation
for the current

$$\frac{V(s)}{R} + \frac{1}{L} \frac{V(s)}{s} + C[sV(s) - v(0^-)] = I_{dc} \left(\frac{1}{s} \right),$$

s-domain equation
for the current

To find $v(t)$: Inverse Transform

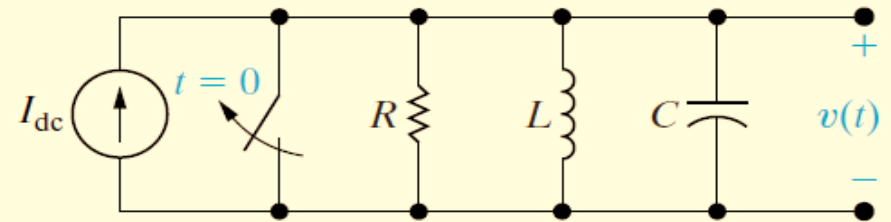
$$V(s) \left(\frac{1}{R} + \frac{1}{sL} + sC \right) = \frac{I_{dc}}{s},$$



$$V(s) = \frac{I_{dc}/C}{s^2 + (1/RC)s + (1/LC)}.$$



$$v(t) = \mathcal{L}^{-1}\{V(s)\}.$$



→ 2nd order
polynomial in s

To find $v(t)$ we must inverse-transform the expression for $V(s)$

To find $v(t)$: Inverse Transform

$$f(t) = \frac{1}{2\pi i} \int_{\alpha-i\infty}^{\alpha+i\infty} F(s) e^{st} ds$$

$$= \sum_k \operatorname{Res}_{s=s_k} \{ F(s) e^{st} \}$$

Sum of residues

To inverse transform we need to decompose the function into **simpler components** that have a straightforward Laplace inverse transform. We use for that *partial fraction expansion*

Fractional Polynomials

- To find the inverse Laplace Transform we express the function $F(s)=N(s)/D(s)$ as a ratio of polynomials: a rational function of “s”

$$F(s) = \frac{N(s)}{D(s)} = \frac{a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}.$$

Simplified notations:

Lowercase time domain

Uppercase s-domain

$$\mathcal{L}\{v\} = V \quad \text{or} \quad v = \mathcal{L}^{-1}\{V\},$$

$$\mathcal{L}\{i\} = I \quad \text{or} \quad i = \mathcal{L}^{-1}\{I\},$$

$$\mathcal{L}\{f\} = F \quad \text{or} \quad f = \mathcal{L}^{-1}\{F\},$$

Inverse Transform

Fractional Polynomials

$$F(s) = \frac{N(s)}{D(s)} = \frac{a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}.$$

Only a *proper rational function* ($m > n$) can be expanded as a sum of partial fractions.

- Proper rational function $n < m$
- Improper rational function $n > m$

Partial Fraction Expansion

Partial Fraction Expansion I

$$F(s) = \frac{N(s)}{D(s)} = \frac{a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}.$$

A rational function can also be expressed as the ratio of two factored polynomials.

$$F(s) = \frac{K(s - z_1)(s - z_2) \cdots (s - z_n)}{(s - p_1)(s - p_2) \cdots (s - p_m)}, m > n$$

$$K = \frac{a_n}{b_m}$$

$z_1, z_2, \dots, z_n$ are the zeros of the system, where $N(z_k) = 0$

$p_1, p_2, \dots, p_m$ are the poles of the system, where $D(p_k) = 0$

Partial Fraction Expansion II

$$F(s) = \frac{K(s - z_1)(s - z_2) \cdots (s - z_n)}{(s - p_1)(s - p_2) \cdots (s - p_m)}, m > n$$

***A proper rational function $F(s)$
can be expressed by:***

$$F(s) = \frac{K_1}{s - p_1} + \frac{K_2}{s - p_2} + \cdots + \frac{K_m}{s - p_m}$$

- The problem is reduced to finding $K_1, K_2, \dots, K_m$

Partial Fraction Expansion III

- Find the parameter K_1

$$F(s) = \frac{K_1}{s - p_1} + \frac{K_2}{s - p_2} + \dots + \frac{K_m}{s - p_m}$$

To find K_1 multiply both sides with $(s - p_1)$, and use the below limit

$$K_1 = \lim_{s \rightarrow p_1} \{ (s - p_1) F(s) \}$$

- $K_2, \dots, K_m$ can be found in a similar way

Distinct Complex Roots of D(s)

Example:

Review

$$F(s) = \frac{100(s + 3)}{(s + 6)(s^2 + 6s + 25)}.$$

p.445 example

$$s^2 + 6s + 25 = (s + 3 - j4)(s + 3 + j4).$$

Hand notes

$$\frac{100(s + 3)}{(s + 6)(s^2 + 6s + 25)} \equiv$$

$$\frac{K_1}{s + 6} + \frac{K_2}{s + 3 - j4} + \frac{K_3}{s + 3 + j4}.$$

$$K_1 = \lim_{s=-6} \{ (s+6) F(s) \}$$

$$K_1 = \frac{100(s+3)}{s^2 + 6s + 25} \Big|_{s=-6} = \frac{100(-3)}{25} = -12,$$

$$K_2 = \lim_{s=-3+j4} \{ (s+3-j4) F(s) \}$$

$$K_2 = \frac{100(s+3)}{(s+6)(s+3+j4)} \Big|_{s=-3+j4} = \frac{100(j4)}{(3+j4)(j8)}$$

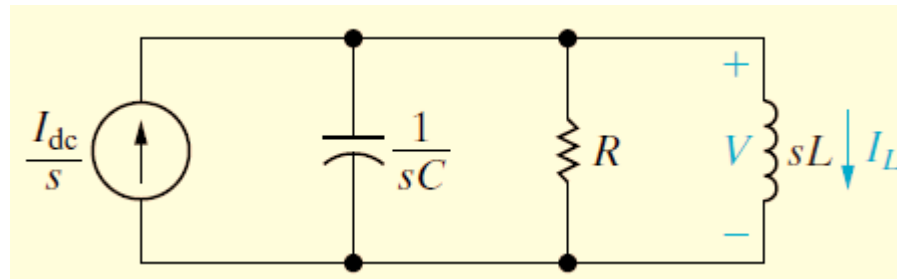
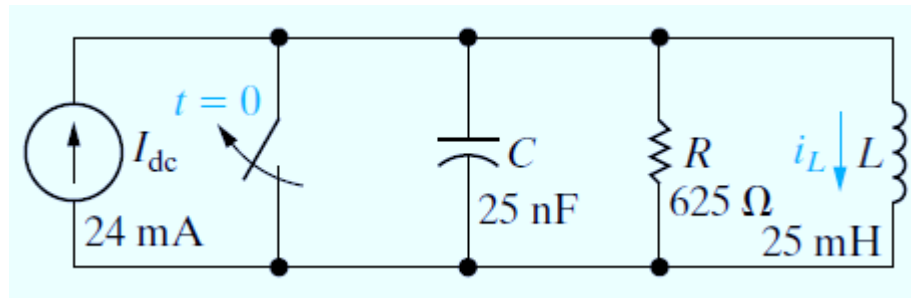
$$= 6 - j8 = 10e^{-j53.13^\circ}$$

$$K_3 = \frac{100(s+3)}{(s+6)(s+3-j4)} \Big|_{s=-3-j4} = \frac{100(-j4)}{(3-j4)(-j8)}$$

$$= 6 + j8 = 10e^{j53.13^\circ}$$

complete conjugate

Going back to the parallel RLC circuit

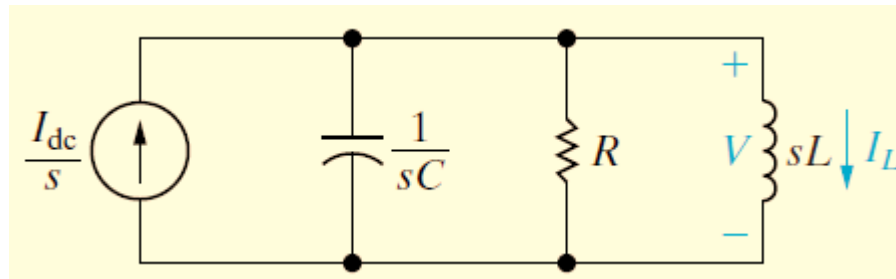
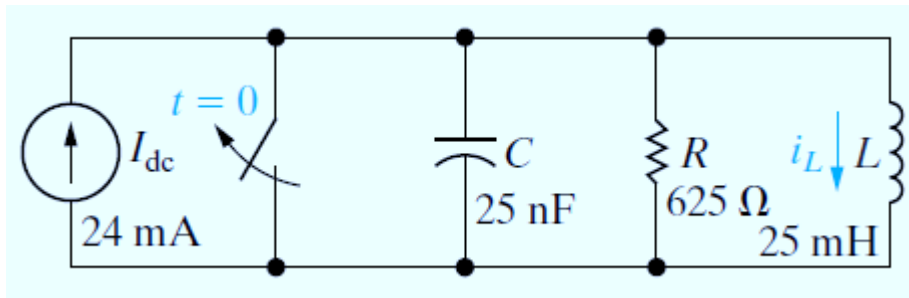


$$V = \frac{I_{dc}/C}{s^2 + (1/RC)s + (1/LC)}.$$

Solved for
the voltage

Step response to a *constant-value input*

p. 471 with example



$$I_L = \frac{I_{dc}/LC}{s[s^2 + (1/RC)s + (1/LC)]}$$

Solving for the voltage V: sum of all currents equals the source current at $t > 0$

$$V = \frac{I_{dc}/C}{s^2 + (1/RC)s + (1/LC)}$$



$$I_L = \frac{V}{sL}$$

Calculating the current in the inductor

Substituting the numerical values

$$I_L = \frac{384 \times 10^5}{s(s^2 + 64,000s + 16 \times 10^8)}.$$

After inserting numerical values

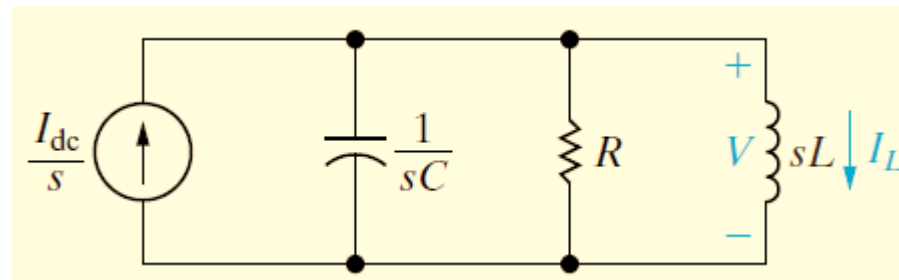
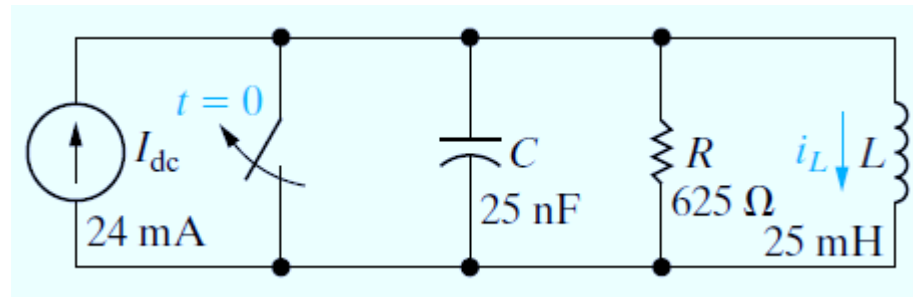
$$I_L = \frac{384 \times 10^5}{s(s + 32,000 - j24,000)(s + 32,000 + j24,000)}.$$

$$I_L = \frac{K_1}{s} + \frac{K_2}{s + 32,000 - j24,000} + \frac{K_2^*}{s + 32,000 + j24,000}.$$

Partial Fraction Expansion

Hand notes

Incorporating the numerical values



$$I_L = \frac{K_1}{s} + \frac{K_2}{s + 32,000 - j24,000} + \frac{K_2^*}{s + 32,000 + j24,000}.$$

Partial fraction expansion: distinct complex roots

Hand notes

$$K_1 = \frac{384 \times 10^5}{16 \times 10^8} = 24 \times 10^{-3},$$

$$K_2 = \frac{384 \times 10^5}{(-32,000 + j24,000)(j48,000)} \\ = 20 \times 10^{-3} \angle 126.87^\circ.$$

$$\mathcal{L}^{-1} \left\{ \frac{K}{s + \alpha - j\beta} + \frac{K^*}{s + \alpha + j\beta} \right\} \\ = 2|K|e^{-\alpha t} \cos(\beta t + \theta).$$

Finding the inverse Laplace

$$i_L = [24 + 40e^{-32,000t} \cos(24,000t + 126.87^\circ)]u(t)\text{mA}.$$

$$I_L = \frac{K_1}{s} + \frac{K_2}{s + 32,000 - j24,000} + \frac{K_2^*}{s + 32,000 + j24,000}.$$

Distinct Complex Roots

$$I_L = \frac{384 \times 10^5}{s(s^2 + 64,000s + 16 \times 10^8)}.$$

$$K_1 = \frac{384 \times 10^5}{16 \times 10^8} = 24 \times 10^{-3},$$

$$K_2 = \frac{384 \times 10^5}{((-32,000 + j24,000)(j48,000))} = 20 \times 10^{-3} \angle 126.87^\circ.$$

Finding the inverse Laplace

$$i_L = [24 + 40e^{-32,000t} \cos(24,000t + 126.87^\circ)]u(t) \text{ mA}.$$

$$\mathcal{L}^{-1} \left\{ \frac{K}{s + \alpha - j\beta} + \frac{K^*}{s + \alpha + j\beta} \right\} = 2|K|e^{-\alpha t} \cos(\beta t + \theta).$$

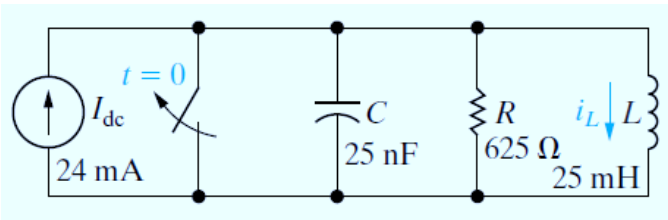
$$K = |K|e^{j\theta} = |K| \angle \theta^\circ,$$

$$K^* = |K|e^{-j\theta} = |K| \angle -\theta^\circ.$$

Equivalent to the solution presented to the same problem in chapter 8: Example 8.7

$$40 \cos(24,000t + 126.87^\circ) = -24 \cos 24,000t - 32 \sin 24,000t.$$

Step response of a parallel RLC circuit- *sinusoidal input*



$$i_g = I_m \cos \omega t \text{ A,}$$

If the *dc* current is replaced with a sinusoidal current

$$I_g = \frac{s I_m}{s^2 + \omega^2}.$$

In the «s» domain

$$V = \frac{(I_g/C)s}{s^2 + (1/RC)s + (1/LC)}.$$

Voltage accross the parallel elements

$$V = \frac{(I_m/C)s^2}{(s^2 + \omega^2)[s^2 + (1/RC)s + (1/LC)]},$$

Calculation of the response

$$I_L = \frac{V}{sL} = \frac{(I_m/LC)s}{(s^2 + \omega^2)[s^2 + (1/RC)s + (1/LC)]}.$$

Solving for the inductor current

$$I_L = \frac{384 \times 10^5 s}{(s^2 + 16 \times 10^8)(s^2 + 64,000s + 16 \times 10^8)}.$$

$$I_L = \frac{384 \times 10^5 s}{(s - j\omega)(s + j\omega)(s + \alpha - j\beta)(s + \alpha + j\beta)},$$

Partial Fraction Expansion
with the complex roots

$$I_L = \frac{K_1}{s - j40,000} + \frac{K_1^*}{s + j40,000} + \frac{K_2}{s + 32,000 - j24,000} + \frac{K_2^*}{s + 32,000 + j24,000}.$$

K₁ & K₂

$$K_1 = \frac{384 \times 10^5(j40,000)}{(j80,000)(32,000 + j16,000)(32,000 + j64,000)}$$

$$= 7.5 \times 10^{-3} \angle -90^\circ, \quad \rightarrow K_1$$

Solved according to rules explained in slide 40

$$K_1 = \lim_{s=j40000} \left\{ I(s) (s - j40000) \right\}$$

$$K_2 = \frac{384 \times 10^5(-32,000 + j24,000)}{(-32,000 - j16,000)(-32,000 + j64,000)(j48,000)}$$

$$= 12.5 \times 10^{-3} \angle 90^\circ, \quad \rightarrow K_2$$

Partial Fraction Expansion

Substituting the numerical values

$$\begin{aligned} i_L &= [15 \cos(40,000t - 90^\circ) \\ &\quad + 25e^{-32,000t} \cos(24,000t + 90^\circ)] \text{ mA}, \\ &= (15 \sin 40,000t - 25e^{-32,000t} \sin 24,000t)u(t) \text{ mA}. \end{aligned}$$

$$\begin{aligned} \mathcal{L}^{-1} \left\{ \frac{K}{s + \alpha - j\beta} + \frac{K^*}{s + \alpha + j\beta} \right\} \\ = 2|K|e^{-\alpha t} \cos(\beta t + \theta). \end{aligned}$$

$$K = |K|e^{j\theta} = |K| \angle \theta^\circ,$$

$$K^* = |K|e^{-j\theta} = |K| \angle -\theta^\circ.$$

$$i_{L_{ss}} = 15 \sin 40,000t \text{ mA},$$

Initial and Steady state
value verification: $t > 0$
and t after a long time

Steady-state current

Other Examples (repeated real roots)

$$\frac{s+6}{s(s+3)(s+1)^2}$$

Proper rational function

$$\frac{s+6}{s(s+3)(s+1)^2} \equiv \frac{K_1}{s} + \frac{K_2}{s+3} + \frac{K_3}{(s+1)^2} + \frac{K_4}{s+1}.$$

Nilsson: p.442

$$\begin{aligned} & \mathcal{L}^{-1} \left\{ \frac{s+6}{s(s+3)(s+1)^2} \right\} \\ &= (K_1 + K_2 e^{-3t} + K_3 t e^{-t} + K_4 e^{-t}) u(t). \end{aligned}$$

Hand notes (p447)

More Examples

For $F(s)$ find the Laplace Inverse

$$F(s) = \frac{96(s + 5)(s + 12)}{s(s + 8)(s + 6)}$$

Nilsson: p.443

Solution

$$\mathcal{L}^{-1} \left\{ \frac{96(s + 5)(s + 12)}{s(s + 8)(s + 6)} \right\} = (120 + 48e^{-6t} - 72e^{-8t})u(t).$$

Simplifying Rules

Review

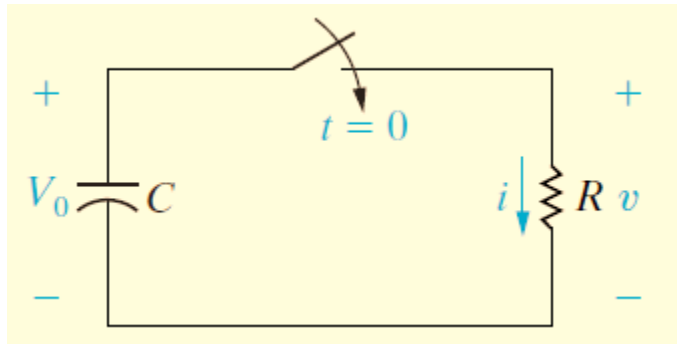
Pair Number	Nature of Roots	$F(s)$	$f(t)$
1	Distinct real	$\frac{K}{s + a}$	$Ke^{-at}u(t)$
2	Repeated real	$\frac{K}{(s + a)^2}$	$Kte^{-at}u(t)$
3	Distinct complex	$\frac{K}{s + \alpha - j\beta} + \frac{K^*}{s + \alpha + j\beta}$	$2 K e^{-\alpha t} \cos(\beta t + \theta)u(t)$
4	Repeated complex	$\frac{K}{(s + \alpha - j\beta)^2} + \frac{K^*}{(s + \alpha + j\beta)^2}$	$2t K e^{-\alpha t} \cos(\beta t + \theta)u(t)$

keypoint

What we know:

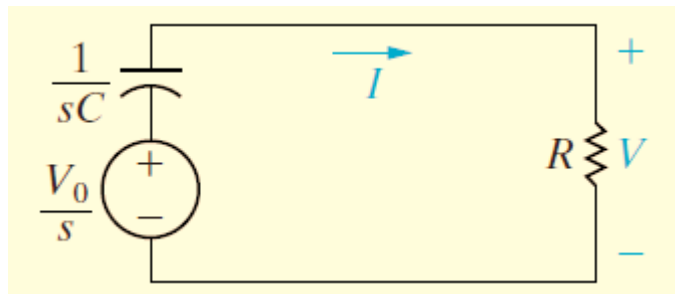
- **How to transform a time-domain function into *s*-domain:** *through Laplace transform;*
- **How to do the calculation in *s*-domain:** *basically, algebraic operations;*
- **How to inverse the *s*-domain result in a simpler way:** *by using the partial fraction expansion.*

Natural response of an RC circuit



Solving for the current I

$$\frac{V_0}{s} = \frac{1}{sC}I + RI.$$



$$I = \frac{CV_0}{RCs + 1} = \frac{V_0/R}{s + (1/RC)}.$$

Proper rational function of s that can easily be inverse transformed

$$v = Ri = V_0 e^{-t/RC} u(t).$$

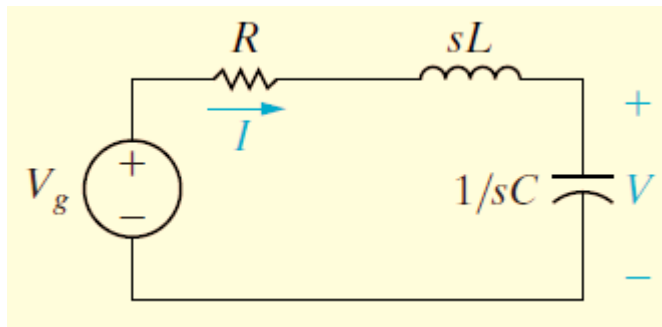
$$i = \frac{V_0}{R} e^{-t/RC} u(t),$$

Break

Transfer Function

The Transfer Function

The **transfer function** is defined as the s-domain ratio of the Laplace transform of the **output (response)** to the Laplace transform of the **input (source)**. In computing the transfer function, we restrict our attention to circuits where all initial conditions are zero.



$$H(s) = \frac{Y(s)}{X(s)},$$

Output response

Input source

Transfer Function (RLC circuit)

$$V_g = I \left(R + sL + \frac{1}{sC} \right)$$

Hand notes

$$H(s) = \frac{I}{V_g} = \frac{1}{R + sL + 1/sC} = \frac{sC}{s^2LC + RCs + 1}.$$

If the current in the capacitor is defined as the output response

$$H(s) = \frac{V}{V_g} = \frac{1/sC}{R + sL + 1/sC} = \frac{1}{s^2LC + RCs + 1}.$$

If the voltage across the capacitor is defined as the output response

The **output signal of interest can vary**: a single circuit can generate **many transfer functions**. When multiple sources are involved, no single transfer function can represent the total output—transfer functions associated with each source must be combined **using superposition** to yield the total response.

The Transfer Function and the Steady-State Sinusoidal Response

Hand notes

✓ Input

$$x(t) = A \cos(\omega t + \phi),$$

$$Y(s) = H(s) \frac{A(s \cos \phi - \omega \sin \phi)}{s^2 + \omega^2}.$$

$$x(t) = A \cos \omega t \cos \phi - A \sin \omega t \sin \phi,$$



$$\begin{aligned} X(s) &= \frac{(A \cos \phi)s}{s^2 + \omega^2} - \frac{(A \sin \phi)\omega}{s^2 + \omega^2} \\ &= \frac{A(s \cos \phi - \omega \sin \phi)}{s^2 + \omega^2}. \end{aligned}$$



Steady state

$$Y(s) = \frac{K_1}{s - j\omega} + \frac{K_1^*}{s + j\omega}$$

+ $\sum$ terms generated by the poles of $H(s)$.

Transient response

Steady-state response

$$Y(s) = H(s) \frac{A(s \cos \phi - \omega \sin \phi)}{s^2 + \omega^2}.$$

$$\begin{aligned} K_1 &= \left. \frac{H(s)A(s \cos \phi - \omega \sin \phi)}{s + j\omega} \right|_{s=j\omega} \\ &= \frac{H(j\omega)A(j\omega \cos \phi - \omega \sin \phi)}{2j\omega} \\ &= \frac{H(j\omega)A(\cos \phi + j \sin \phi)}{2} = \frac{1}{2}H(j\omega)Ae^{j\phi}. \end{aligned}$$

$$K_1 = \frac{A}{2} |H(j\omega)| e^{j[\theta(\omega) + \phi]}.$$

$$\begin{aligned} \mathcal{L}^{-1} \left\{ \frac{K}{s + \alpha - j\beta} + \frac{K^*}{s + \alpha + j\beta} \right\} \\ = 2|K|e^{-\alpha t} \cos(\beta t + \theta). \end{aligned}$$

$$H(j\omega) = |H(j\omega)|e^{j\theta(\omega)}.$$

$$y_{ss}(t) = A|H(j\omega)| \cos [\omega t + \phi + \theta(\omega)],$$

$$K = |K|e^{j\theta} = |K| \angle \theta^\circ,$$

affects here!

keypoint

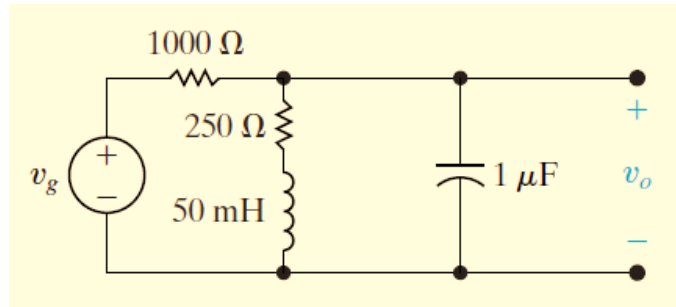
The terms generated by the poles of $H(s)$ do not contribute to the steady state response of $y(t)$ because all these poles lie in the left half of the s plane. This means that the time domain terms approach to zero as t increases

$$Y(s) = \frac{K_1}{s - j\omega} + \frac{K_1^*}{s + j\omega} + \sum \text{terms generated by the poles of } H(s).$$

Therefore, the first two terms on the right-hand side of the above equation determine the steady-state response. The problem is then reduced to finding K_1

Exercise

Example 13.2 (Nilsson)



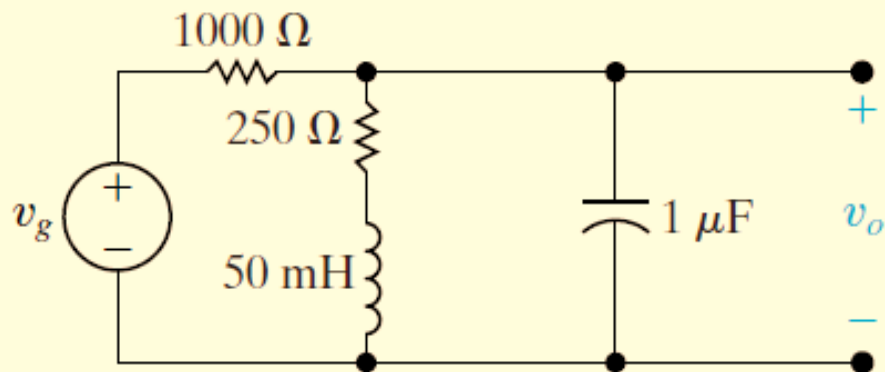
$$v_g = 50tu(t)$$

Same transfer function as example 13.1

- Use the transfer function to find v_o .
- Identify the transient component of the response.
- Identify the steady-state component of the response.

$$H(s) = \frac{V_o}{V_g}$$
$$= \frac{1000(s + 5000)}{s^2 + 6000s + 25 \times 10^6}$$

Example 13.4 (Nilsson)



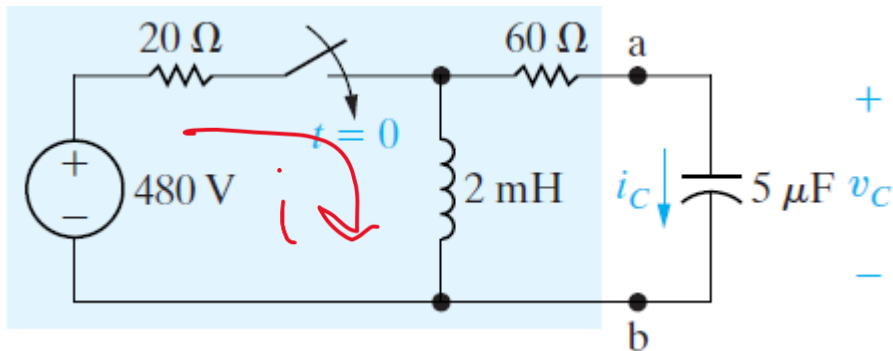
Same transfer function as example 13.1

The sinusoidal source voltage is $120 \cos(5000t + 30^\circ)$ V. Find the steady-state expression for v_o .

$$H(s) = \frac{V_o}{V_g}$$
$$= \frac{1000(s + 5000)}{s^2 + 6000s + 25 \times 10^6}$$

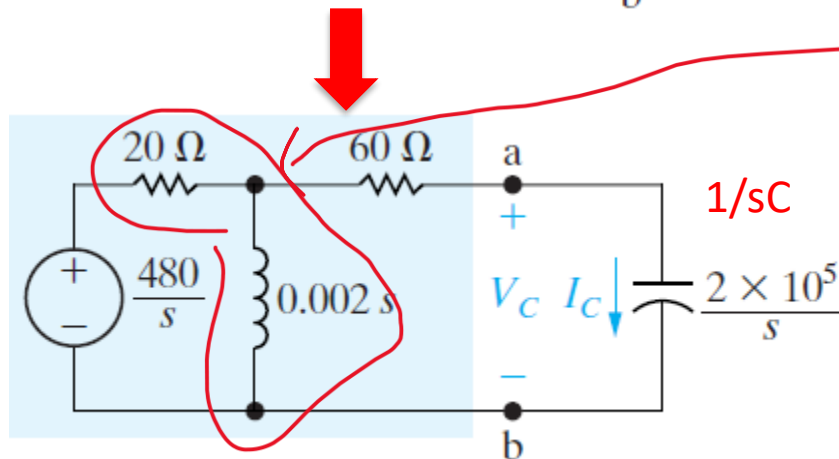
Thevenin Equivalent

The use of Thevenin's Equivalent



Thevenin voltage is the open circuit voltage across terminals a and b

$$V_{Th} = \frac{(480/s)(0.002s)}{20 + 0.002s} = \frac{480}{s + 10^4}$$

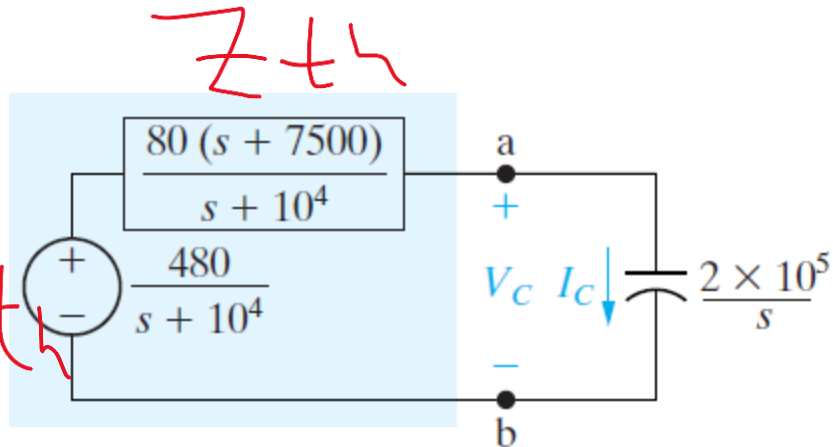


Thevenin impedance seen from a & b is the combination of resistor 60 ohm in series with parallel combination of 20 ohm and 0.002 s

Find the capacitor current that results from closing the switch

$$Z_{Th} = 60 + \frac{0.002s(20)}{20 + 0.002s} = \frac{80(s + 7500)}{s + 10^4}$$

Using Thevenin Equivalent



Capacitor current equals the Thevenin voltage divided by the total series impedance

$$I_C = \frac{480/(s + 10^4)}{[80(s + 7500)/(s + 10^4)] + [(2 \times 10^5)/s]}.$$

Simplifying and using partial fraction expansion

$$I_C = \frac{6s}{s^2 + 10,000s + 25 \times 10^6} = \frac{6s}{(s + 5000)^2}.$$

$$I_C = \frac{-30,000}{(s + 5000)^2} + \frac{6}{s + 5000},$$

Handwritten red arrow from the previous equation to this one.

$$i_C = (-30,000te^{-5000t} + 6e^{-5000t})u(t) \text{ A.}$$

Find initial current and voltage in the capacitor

$$i_C = (-30,000te^{-5000t} + 6e^{-5000t})u(t) \text{ A.}$$

(t=0)

$$i_C(0) = 6 \text{ A.}$$

Using Laplace

$$V_C = \frac{1}{sC} I_C = \frac{2 \times 10^5}{s} \frac{6s}{(s + 5000)^2}$$
$$= \frac{12 \times 10^5}{(s + 5000)^2},$$

$$v_C = 12 \times 10^5 te^{-5000t} u(t)$$

Verify by using time domain approach



$$v_C = 2 \times 10^5 \int_{0^-}^t (6 - 30,000x)e^{-5000x} dx.$$

End